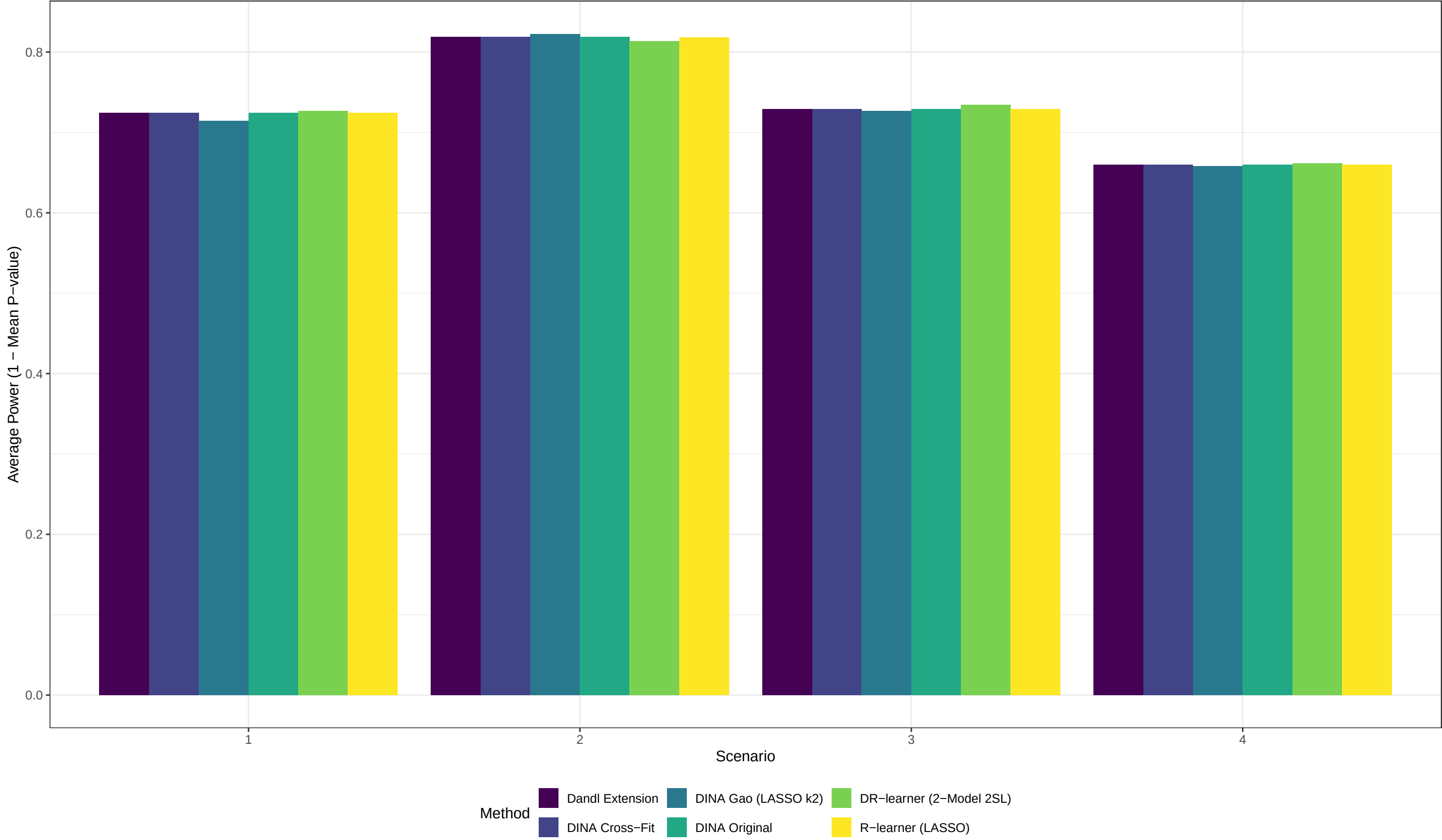
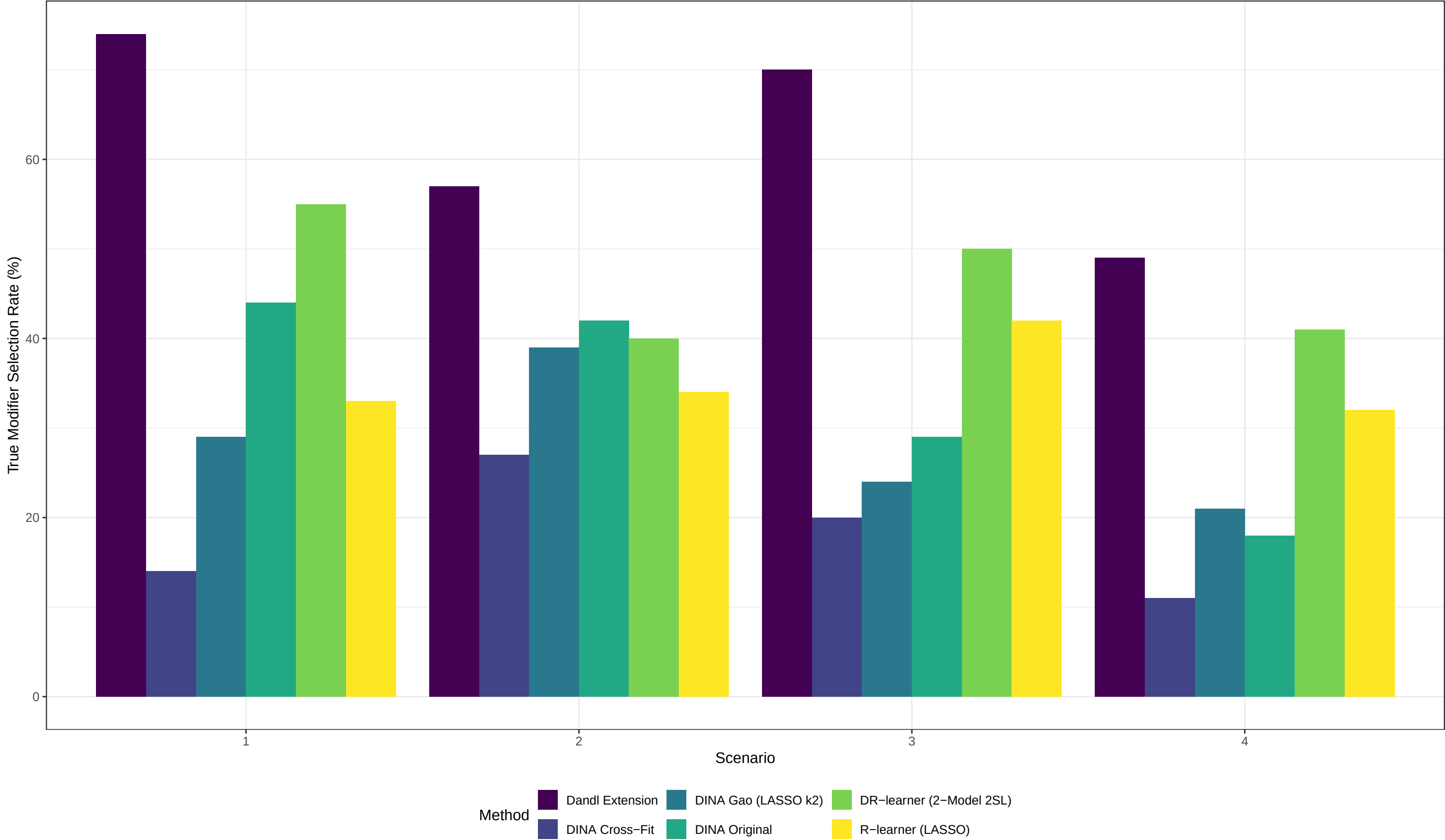


Objective 1: Average Power Across All Scenarios
Higher = better ability to detect heterogeneity



Objective 2: True Modifier Selection Rate Across Scenarios
% of times the top-ranked variable was a true effect modifier



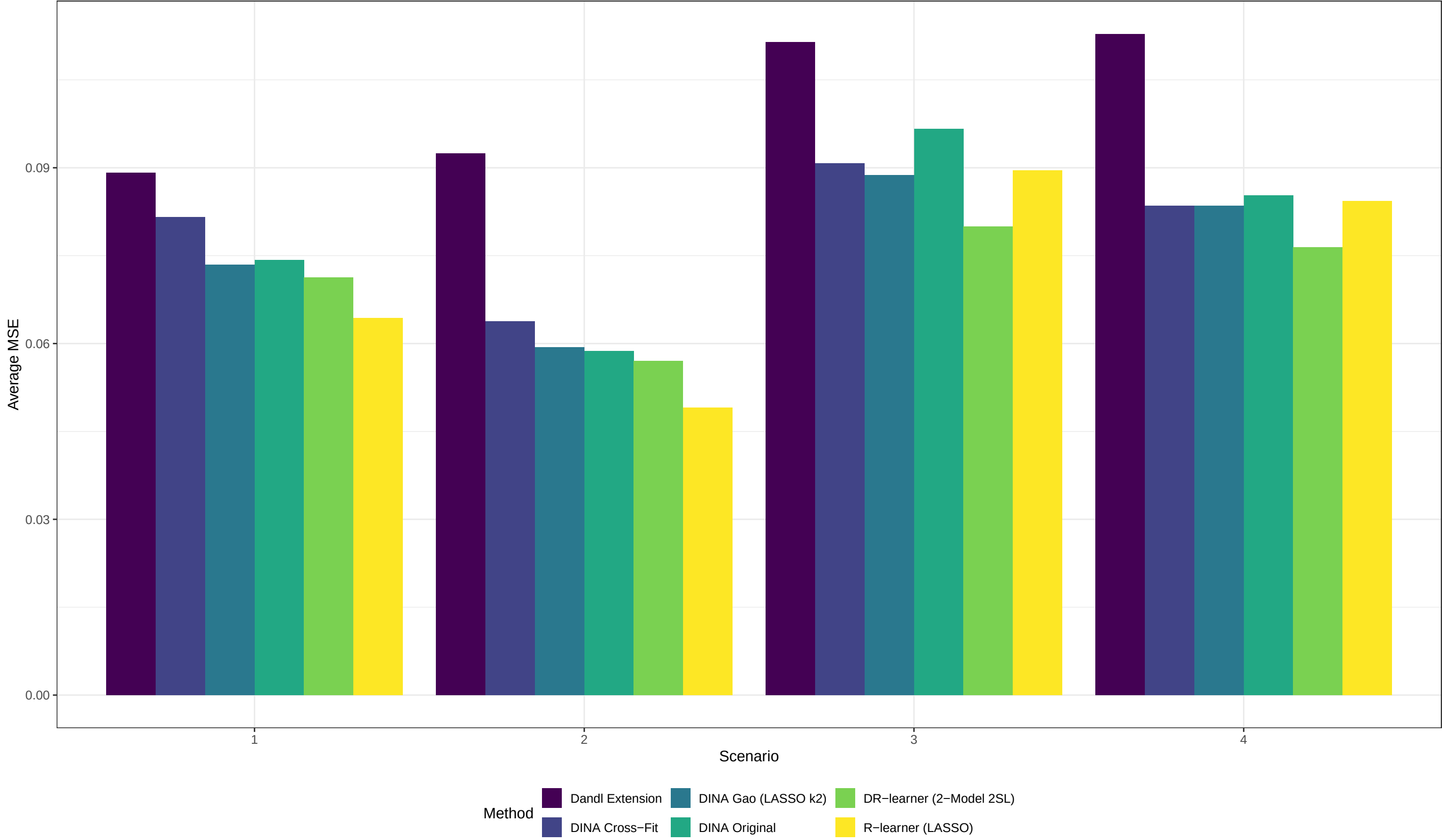
TE-VIM TMLE (Li 2026): Scaled VIM (VIMb) Ranking Across Scenarios (beta>0)

Higher = covariate explains more treatment-effect heterogeneity



Objective 3: Average MSE Across All Scenarios

Lower = better CATE estimation



Overall Method Rankings (All Scenarios, . > 0)

Method	Scenarios	Avg_Power	Avg_Detection_Rate	Avg_MSE	Avg_Correlation	Avg_Somers_D	Avg_True_Selection
DR-learner (2-Model 2SL)	4	0.734	33.25	0.071	0.365	0.269	46.50
Dandl Extension	4	0.733	33.00	0.101	0.474	0.339	62.50
DINA Cross-Fit	4	0.733	33.00	0.080	0.241	0.183	18.00
DINA Original	4	0.733	33.25	0.079	0.427	0.355	33.25
R-learner (LASSO)	4	0.733	33.25	0.072	0.502	0.422	35.25
DINA Gao (LASSO k2)	4	0.730	32.75	0.076	0.335	0.267	28.25

Scenario	True_Modifiers	N_Modifiers	Prognostic_Only
1	X11	1	X1
2	X14	1	X8
3	X14, X1	2	X17
4	X14, X4	2	X11

Wall-clock time per CATE method (mean $\pm$ sd across simulations)

